

HHI_PCB_APD-Array-EVB - Overview

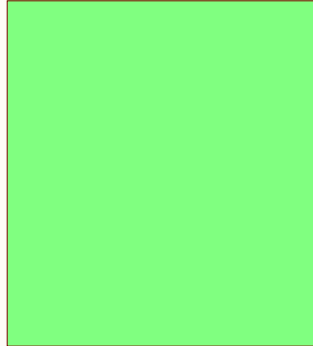
Short Description:

Dev Board PCB for the Broadcom 12x12 APD Array chip in a DDR3 format. The PCB has all power supply components, SMA output for 4 differential outputs, the APD chip array and the microcontroller.

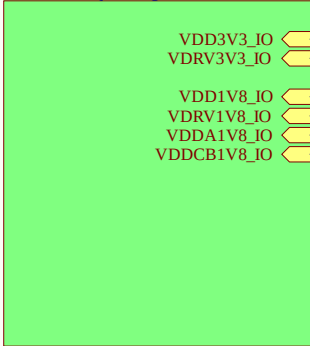
Variant Informations:

Variant:	VAR-01
Variant Name:	Standard

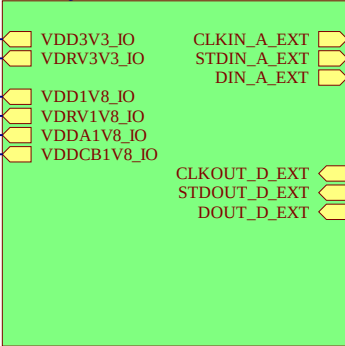
Power
01-Power.SchDoc



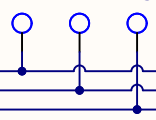
Power-Sequencing
02-Power-Sequencing.SchDoc



Microprocessor
03-Microprocessor.SchDoc



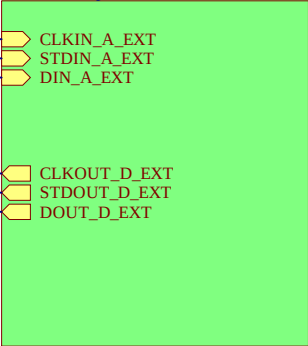
TP1 TP2 TP3



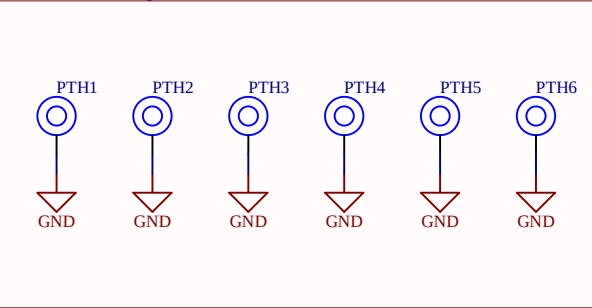
TP4 TP5 TP6



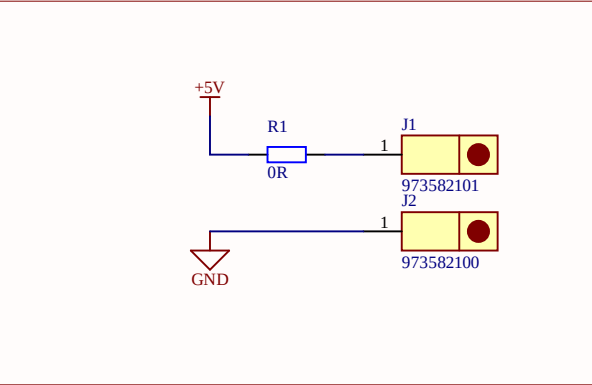
ASIC-Chip
04-ASIC-Chip.SchDoc



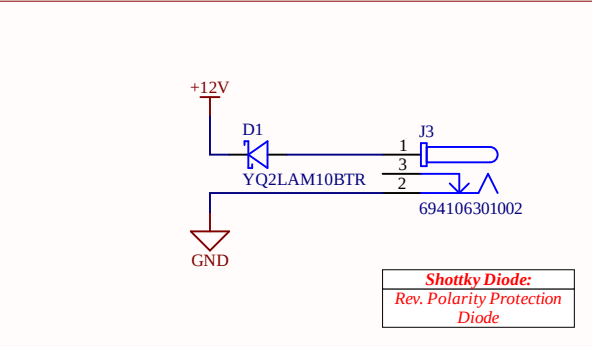
Mechanical Mounting Holes



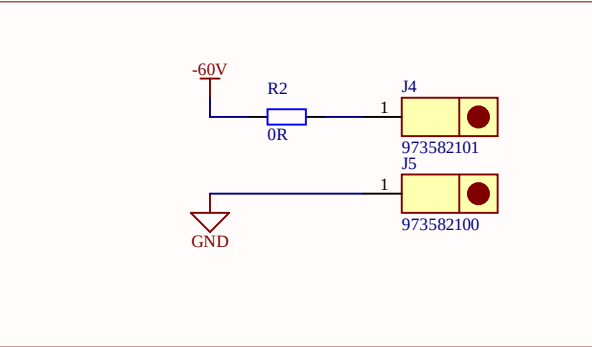
+5V Connector



-12V Connector



-60V Connector

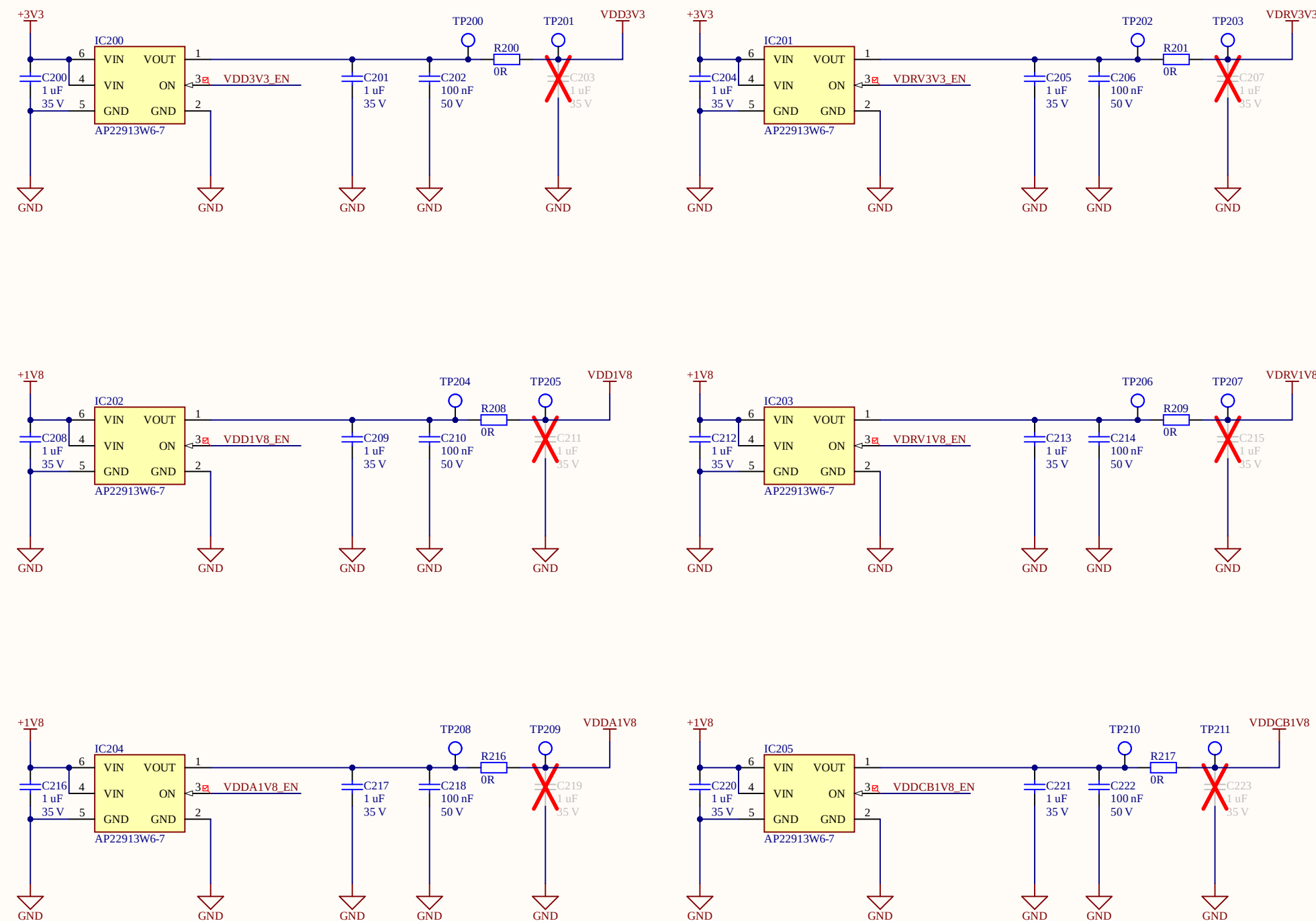


Change Log:

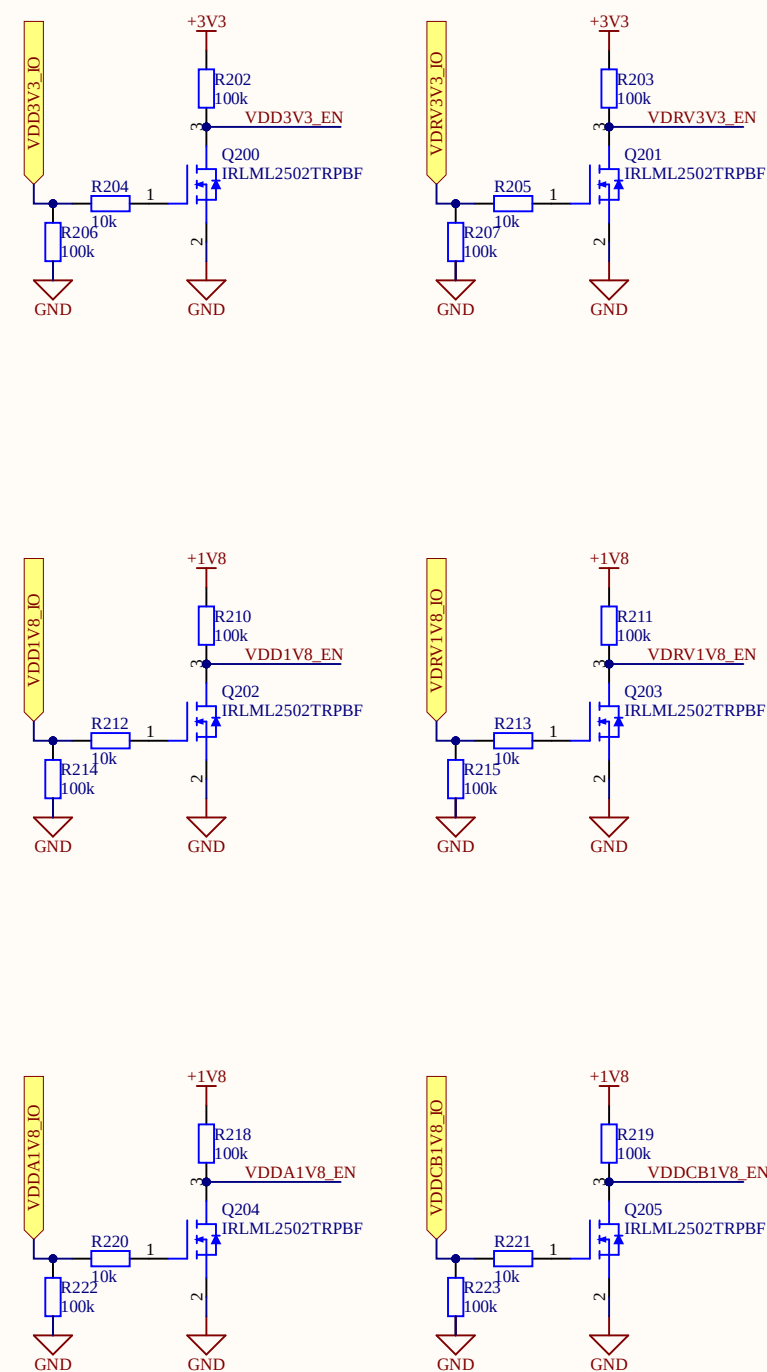
Version	Changes
V0.1.0	initial design
Vx.x.x	changes
Vx.x.x	changes
Vx.x.x	changes

Title: HHI_PCB_APD-Array-EVB				
PCB ID: HHI_PCB_APD-Array-EVB_V0-1-0		08/2025		
Document: 00-Overview.SchDoc		Version: V0.1.0		
Editor: YS	Date: 03.12.2025	Size: A3		
Approver: GB / OL	Time: 16:28:15	Sheet 1 of 4		
File: C:\Users\lubritz\Documents\ temp\HHI_PCB_APD-Array-EVB_V0-1-0\Sheets\00-Overview.SchDoc				

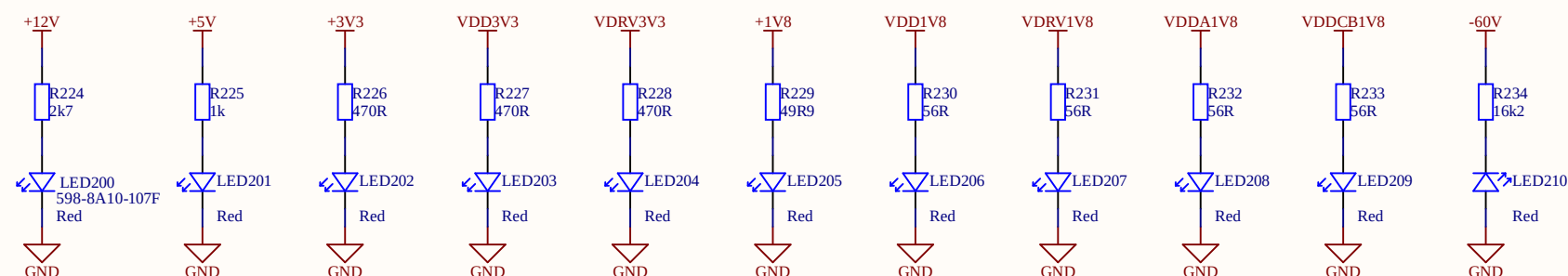
POWER_SEQ -1- Load Switches



POWER_SEQ -2- Enable Logic



POWER_SEQ -3- Status LEDs

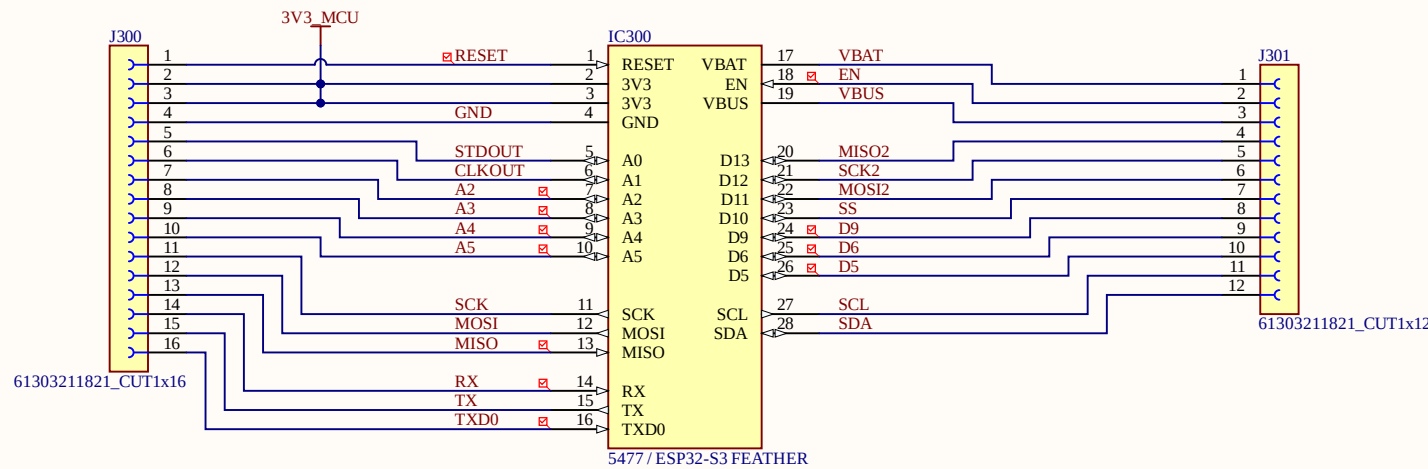


EN Pins:
All Enable Pins pulled up and connected to MCU should have max 3V3!!! MCU Pins are only 3V3 compatible!!

Title: HHI_PCB_APD-Array-EVB			08/2025
PCB ID: HHI_PCB_APD-Array-EVB_V0-1-0			Version: V0.1.0
Document: 02-Power-Sequencing.SchDoc			
Editor: YS	Date: 03.12.2025	Size: A3	
Approver: GB / OL	Time: 16:28:16	Sheet 3 of 4	
File: C:\Users\ubritz\Documents\ temp\HHI_PCB_APD-Array-EVB_V0-1-0\Sheets\02-Power-Sequencing.SchDoc			

uPROCESSOR -1- Microcontroller and Pin Headers

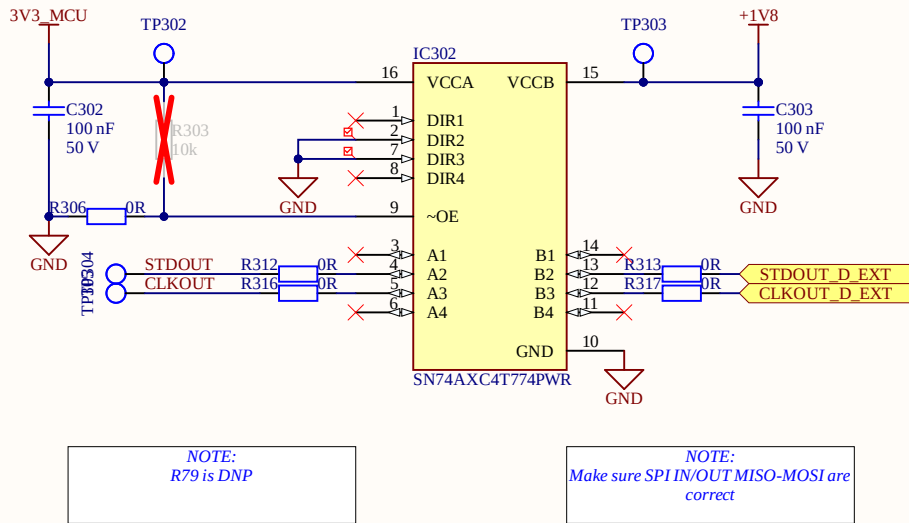
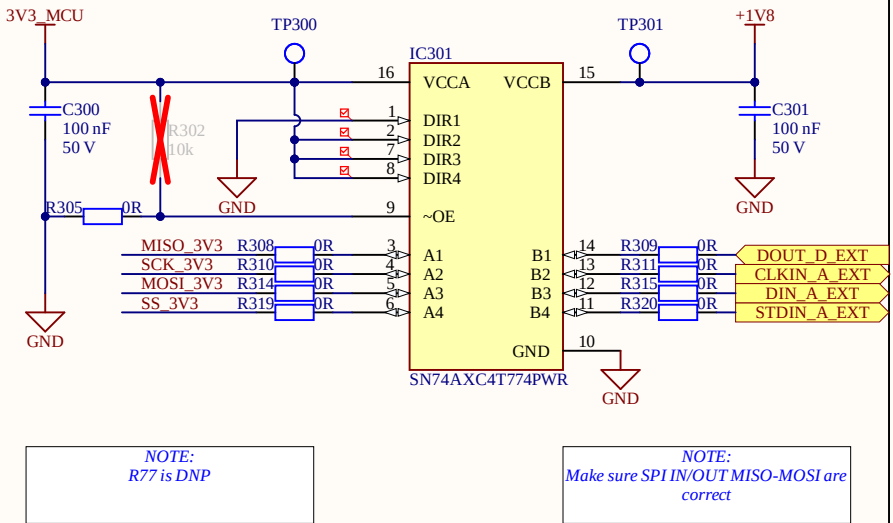
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P32-S3_Pinout-3488814020.png. File does not exist.



uPROCESSOR -2- In/Logic Shifter

uPROCESSOR -3- Out/Logic Shifter

uPROCESSOR -3- SPI and GPIOs



NOTE:
Goes to load
switches enable
pins in Power Seq.

VDD1V8_IO	D9
VDDA1V8_IO	D6
VDDCB1V8_IO	D5

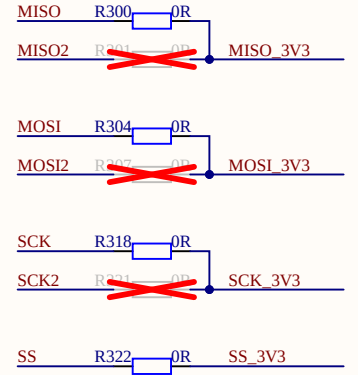
VDRV1V8_IO	A2
VDD3V3_IO	A3
VDRV3V3_IO	A4

NOTE:
Populate only one set of 0R jumpers for 1
SPI Per.

EX: Only MISO, MOSI, SCK, SS

OR

Only MISO2, MOSI2, SCK2, SS

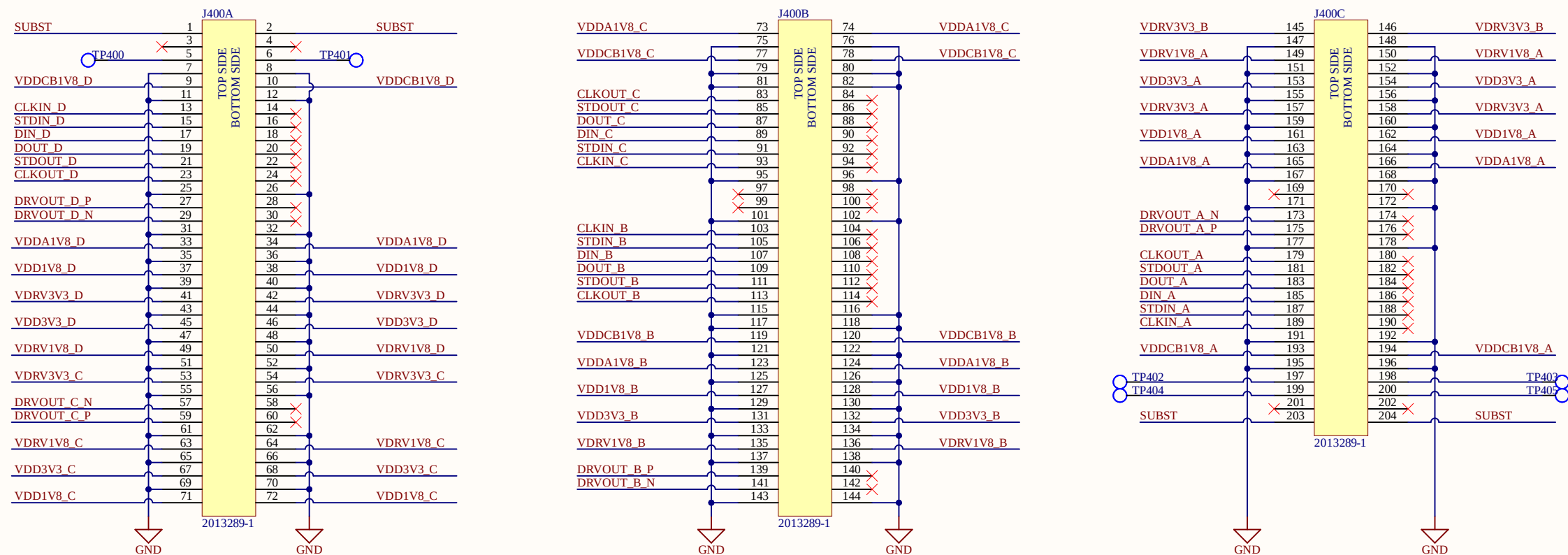


NOTE:
Make sure SPI bus on D13-D10 works or if
we should use SPI bus 2

MOSI - DIN_A // CS - STDIN_A

Add pinheaders

ASIC -1- DDR3 Connector



ASIC -2- Peripherals

